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Comparing Brain Activation Patterns in Stress-Induced and Post-Stress Recovery States of Highly and Moderately Stressed Individuals

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Abstract: Our study aimed to identify the mechanisms of stress-induced and post-stress recovery states by evaluating voxel type and the total number of voxels activated based on Perceived Stress Scale scores. Functional MRI (fMRI) was used to measure the brain activation patterns in participants grouped into moderate- and high-stress categories. The number of activation voxels in the parahippocampal gyrus of the limbic lobe was greater in the high-stress group than in the moderate-stress group. Moreover, only the high-stress group showed the characteristic activation of the left precuneus. The moderate-stress group showed a greater number of activation voxels than the high-stress group for the occipital, frontal, and limbic lobes, while the reverse was true for the parietal lobe. In the post-stress recovery phase, the left lingual, inferior frontal, and middle frontal gyri were activated, and the number of activation voxels in these areas was greater in the high-stress group than in the moderate-stress group. The characteristic activation of the declive in the left cerebellum was observed in the moderate-stress group, whereas the activation of the right cuneus was dominant in the high-stress group. Our findings suggest that perceived stress may play an important role in determining the neural mechanisms underlying stress and post-stress recovery, providing insights into the complex interplay between the psychological and neural processes occurring in response to stress.

Keywords: stress score; stress; post-stress recovery; Perceived Stress Scale (PSS); neural mechanisms

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1. Introduction

Stress is a multifaceted phenomenon with widespread effects on both physical and mental health. Brain activation patterns associated with stress induction often include the amygdala, prefrontal cortex, and anterior cingulate cortex, which play essential roles in emotional processing, attention, and stress regulation [1,2]. Additionally, the hypothalamus–pituitary–adrenal (HPA) axis is activated during stress, leading to the release of cortisol, a key stress hormone [3]. During post-stress recovery, this activation pattern shifts, engaging regions like the ventromedial prefrontal cortex, posterior cingulate cortex, and hippocampus, which are implicated in emotional regulation, cognitive control, and memory consolidation [1,2]. As recovery progresses, the HPA axis gradually returns to baseline, resulting in decreased cortisol levels [3].

Several studies have explored how changes in perceived stress influence brain activation patterns. For instance, individuals with higher chronic stress have been shown to exhibit increased activation in the amygdala and reduced activity in the prefrontal cortex, suggesting alterations in their emotional regulation and cognitive control [4]. Similarly, human studies using interventions like mindfulness have reported a reduction in perceived stress correlating with increased activation in regions such as the anterior cingulate cortex and dorsolateral prefrontal cortex during emotional regulation tasks [5]. These findings

highlight the significant impact of stress on brain function, but a detailed comparison of brain activation patterns based on different stress levels remains underexplored.

The Perceived Stress Scale (PSS) is a widely used self-report questionnaire that measures subjective stress levels, providing a useful tool for categorizing individuals based on their perceived stress. Previous research indicates that higher PSS scores are associated with greater activity in brain regions involved in emotional processing, such as the amygdala and hippocampus, and decreased activity in regions associated with emotion regulation, like the prefrontal cortex [6,7]. A longitudinal study even found that increases in PSS scores over time were linked to increased activation in the amygdala and hippocampus, further emphasizing the relationship between perceived stress and brain function [7]. While these studies shed light on the connection between stress perception and brain activation, systematic research comparing brain patterns in stress-induced and post-stress recovery states based on PSS scores is limited. Moreover, the neural mechanisms involved in resilience during the post-stress recovery state remain unclear.

Understanding the neural mechanisms underlying both stress induction and recovery is crucial for identifying how subjective stress levels impact brain function. Grouping individuals by their PSS scores allows researchers to investigate differences in the brain activation patterns between those with moderate and high stress. By examining these patterns, researchers can identify which brain regions are more active in highly stressed individuals and explore neural correlates of resilience, which could provide insights into interventions for enhancing stress recovery. For example, identifying unique activation patterns in individuals with high versus moderate stress during recovery could highlight neural pathways that facilitate resilience.

Therefore, this study aimed to identify the mechanism of action for each condition by examining the characteristically activated areas of the brain and the number of voxels during stress-induced and post-stress recovery states in moderate- and high-stress groups classified by their PSS scores.

2. Research Methods

2.1. Participants

This study included healthy right-handed male and female participants with normal cognitive function. The Korean version of the Perceived Stress Scale (PSS) test [8] was administered prior to the experiment to classify the participants into moderate- or high-stress groups.

The Perceived Stress Scale (PSS) is a commonly used assessment tool for describing an individual's level of perceived stress. Numerous studies have investigated the PSS in the diagnosis of stress, with overall results suggesting that the scale is a reliable measure of perceived stress with a good internal consistency and test–retest reliability [8,9]. One study found that the PSS had a high internal consistency, with a Cronbach's alpha of 0.84, indicating that the items on the scale measured the same underlying construct of perceived stress [8]. Another study found that the PSS has a good test–retest reliability, with a correlation coefficient of 0.85 over a two-week period [9].

For the PSS test, the total score ranges from 0 to 40, with higher scores indicating a greater level of perceived stress. In our study, the moderate-stress group was defined as having a PSS score ranging from 15 to 18, whereas the high-stress group was defined as having a score of 19 or greater.

The exclusion criteria included individuals with metal objects in their bodies, such as pacemakers or metal implants, due to potential interference during magnetic resonance imaging (MRI). Those with claustrophobia were also excluded, as they might have experienced discomfort during the MRI scans. Additionally, individuals with a history of neurological or psychiatric disorders, including depression, anxiety, or epilepsy, were excluded to avoid confounding factors affecting brain activation. Participants currently taking psychoactive medications or substances that could alter their cognitive or neural responses were not included. Individuals with a history of substance abuse, cardiovascular

diseases—given the potential health risks posed by stress-induced tasks—and those reporting irregular sleep patterns, as sleep quality can significantly impact stress levels and brain function, were also excluded. The protocol and rationale of the study were described to the participants prior to enrolment, and individuals were instructed to refrain from activities that may affect their brain activation, including smoking, alcohol, and coffee. All participants were right-handed, according to the revised Edinburgh Reading Test. All participants gave their written informed consent before any study related procedures were performed.

The protocol for the research project was approved by the Institutional Review Committee of Konkuk University, within which the work undertaken conformed to the provisions of the Declaration of Helsinki (IRB number: 7001355-202010-HR-405).

2.2. Defining the Stress-Inducing Task

The stress-inducing task was performed using the Montreal Imaging Stress Task (MIST) stress paradigm. The MIST [10] was devised to induce psychosocial stress, in which the researcher provides negative feedback on task performance, subjecting participants to continuous negative evaluations from others in an uncontrollable situation. Specifically, it induces social stress by causing the study participants to perceive that their performance is very poor compared to the virtual average performance, irrespective of their actual performance.

To induce stress, elementary arithmetic problems were presented by difficulty over a time limit of 3 s, after which, the participants were given feedback on the answer that they had provided, as well as the mean score. Prior to initiating the test, the participants were asked to maintain an average score of 95.

2.3. Study Design

The study was conducted in five stages. In the training phase, the participants were given time to adapt to the response button and solve problems; in the rest phase (90 s), they were allowed to rest comfortably; in the control phase (90 s), they solved problems without any stress inducers; and in the stress task phase, they solved problems under stress-inducing conditions. To induce acute stress (300 s), a visual of the time limit (3 s) was presented, and the participants were given feedback on the problem they had just solved (correct, incorrect, or no response) for 2 s along with their mean score before moving on to the next question. The number of presented items was 60. The recovery phase (180 s) extracted brain signal changes when the participants were in a stable condition after performing all tasks. Functional MRI (fMRI) images were acquired for all participants throughout the entire duration of the experimental phases.

2.4. Functional MRI (fMRI) Measurements

Images were acquired using a 3T MRI system (Magnetom TrioTim; Siemens Medical Systems, Erlangen, Germany) with a standard 16-channel head coil. Single-shot echo-planar fMRI scans were obtained in 29 continuous slices parallel to the anterior–posterior commissure line. The fMRI parameters were as follows: repetition time/echo time, 2000/20 ms; FOV, 240 mm; flip angle, 77°; matrix, 128 × 128; slice thickness, 3 mm; and voxel size, 3.0 mm × 3.0 mm × 3.0 mm. Anatomical images were obtained using a T1-weighted 3D MPRAGE sequence with the following parameters: TR/TE, 1900/2.52 ms; FOV, 256 mm; flip angle, 9°; matrix, 256 × 256; slice thickness, 1 mm; and voxel size, 1.0 mm × 1.0 mm × 1.0 mm.

2.5. Imaging Analysis of Brain Function

fMRI data were analyzed using SPM 12 software (Wellcome Department of Cognitive Neurology, London, UK). All functional images were aligned with the anatomical images using affine transformation routines built into the SPM 12 program. A time series of images acquired from the same participants was determined using a least-squares approach and a 6-parameter (rigid body) spatial transformation. The first image in the list specified

by the user was utilized as the reference to which all subsequent scans were realigned. The realigned scans were co-registered with the participants' anatomic images obtained during each session and normalized to a template image in SPM 12 (using the space defined by the Montreal Neurologic Institute). Motion correction was performed using sinc interpolation. Time series data were filtered with a 240 s high-pass filter to remove artifacts due to cardiorespiratory and other cyclical influences. Moreover, co-registered T1 images were used via multichannel segmentation to extract probabilistic maps of the six following tissue classes: GM, WM, CSF, bone, soft tissue, and residual noise. The functional map was smoothed using an 8 mm isotropic Gaussian kernel prior to statistical analysis. Statistical analyses were performed at the group level using a general linear model and the theory of Gaussian random fields implemented in SPM 12. Group analyses were performed to extend the inferences of individual activation to the general population, from which the participants were drawn to list all clusters above the chosen level of significance, as well as the separate (>8 mm apart) maxima within each cluster, with the details of significance thresholds ($p < 0.05$) and search volume presented underneath.

Using the subtraction method, the areas of the brain activated under stress-inducing conditions in the moderate- and high-stress groups were identified and compared to the regions of the brain activated during the post-stress recovery state. Additionally, the number of voxels activated in each lobe during the stress and recovery phases was determined, and the number of activated voxels between the two groups was compared by phase using an independent *t*-test.

3. Results

A total of 37 participants (24.4 ± 4.3 years) were included in the moderate-stress group, comprising 17 males (25.1 ± 3.0 years) and 20 females (24.6 ± 4.2 years) whose age ranged from 20 to 33 years old. The high-stress group comprised a total of 38 participants (26.2 ± 4.1 years), including 15 males (26.5 ± 4.4 years) and 23 females (26.0 ± 3.9 years) whose age ranged from 20 to 34 years old.

Tables 1 and 2 present the brain regions activated during the stress-induced and post-stress recovery states for the moderate-stress and high-stress groups, respectively. Also, Figure 1 displays the brain mapping images of both the moderate-stress and high-stress groups during the stress-inducing condition, while Figure 2 shows the brain mapping images of the two groups during the recovery state. Within the moderate-stress group, the middle temporal gyrus, superior occipital gyrus, superior frontal gyrus, and anterior cingulate were primarily activated under stress-inducing conditions (Table 1), whereas precuneus activation was predominant in the high-stress group (Table 2). The total number of activated voxels was higher in the moderate-stress group than in the high-stress group.

During the post-stress recovery phase, the moderate-stress group showed the activation of the inferior occipital gyrus, lingual gyrus, and declive, whereas the high-stress group showed the strong activation of the cuneus, lingual gyrus, and inferior occipital gyrus. The total number of activated voxels was higher in the high-stress group than in the moderate-stress group.

Furthermore, the number of voxels activated in each activated area of each lobe was identified for each phase (stress and recovery) in the moderate- and high-stress groups for comparison (Figure 3a,b). In the stress phase, the number of activation voxels was greater in the occipital, frontal, and limbic lobes in the moderate-stress group compared to the high-stress group. In contrast, the parietal lobe showed a greater number of activation voxels in the high-stress group compared to the moderate-stress group (Figure 3a). During the recovery phase, the number of activation voxels was greater in the occipital and frontal lobes in the high-stress group compared to the moderate-stress group. In contrast, cerebellar activation was predominant in the moderate-stress group (Figure 3b).

Table 1. Regions activated in post-stress recovery state in the moderate-stress group (control < stress, FWE 0.05).

Phase	Voxel	T-Value	x	y	z	Side	Lobe	Region	Brodmann Area
Stress	432	3.80	−51	−61	17	Left Cerebrum	Occipital Lobe	Middle Temporal Gyrus	BA 19
	400	3.52	−42	−79	26	Left Cerebrum	Occipital Lobe	Superior Occipital Gyrus	BA 19
	310	3.78	−12	59	17	Left Cerebrum	Frontal Lobe	Superior Frontal Gyrus	
	275	3.37	−15	38	5	Left Cerebrum	Limbic Lobe	Anterior Cingulate	
	49	3.44	−51	35	2	Left Cerebrum	Frontal Lobe	Inferior Frontal Gyrus	BA 45
	34	2.25	0	8	−7	Left Cerebrum	Limbic Lobe	Anterior Cingulate	
	23	2.96	−3	53	−7	Left Cerebrum	Frontal Lobe	Medial Frontal Gyrus	BA 10
	16	2.62	−24	−49	11	Left Cerebrum	Sub-lobar	Lateral Ventricle	
	8	1.85	−21	−43	5	Left Cerebrum	Limbic Lobe	Parahippocampal Gyrus	
	23	2.23	−27	14	23	Left Cerebrum	Frontal Lobe	Sub-Gyral	
	12	2.21	−18	−49	62	Left Cerebrum	Parietal Lobe	Superior Parietal Lobule	BA 7
	14	2.18	−51	−4	−4	Left Cerebrum	Temporal Lobe	Superior Temporal Gyrus	
	35	2.52	−3	−13	50	Left Cerebrum	Frontal Lobe	Medial Frontal Gyrus	BA 6
	37	2.57	−18	−37	20	Left Cerebrum	Sub-lobar	Lateral Ventricle	
	10	2.32	−21	5	−22	Left Cerebrum	Limbic Lobe	Uncus	BA 28
	92	3.76	−9	−55	29	Left Cerebrum	Limbic Lobe	Cingulate Gyrus	BA 31
	68	2.76	3	−58	32	Right Cerebrum	Parietal Lobe	Precuneus	BA 7
	59	2.40	6	−52	23	Right Cerebrum	Limbic Lobe	Posterior Cingulate	BA 31
	215	3.65	57	−1	35	Right Cerebrum	Frontal Lobe	Precentral Gyrus	
	177	3.00	60	−13	8	Right Cerebrum	Temporal Lobe	Superior Temporal Gyrus	
	58	3.49	54	−64	23	Right Cerebrum	Temporal Lobe	Middle Temporal Gyrus	BA 39
	44	2.68	42	−79	26	Right Cerebrum	Occipital Lobe	Superior Occipital Gyrus	BA 19
	26	2.53	12	47	41	Right Cerebrum	Frontal Lobe	Superior Frontal Gyrus	
	21	2.40	33	−73	5	Right Cerebrum	Occipital Lobe	Middle Occipital Gyrus	
	23	2.25	24	−40	14	Right Cerebrum	Sub-lobar	Extra-Nuclear	
	10	2.17	27	−4	35	Right Cerebrum	Frontal Lobe	Sub-Gyral	
Recovery	780	12.69	−27	−91	−10	Left Cerebrum	Occipital Lobe	Inferior Occipital Gyrus	
	740	12.04	−9	−91	−4	Left Cerebrum	Occipital Lobe	Lingual Gyrus	
	693	11.27	−24	−82	−19	Left Cerebellum	Posterior Lobe	Declive	
	159	8.88	−27	−61	44	Left Cerebrum	Parietal Lobe	Superior Parietal Lobule	
	100	5.67	−51	−37	47	Left Cerebrum	Parietal Lobe	Inferior Parietal Lobule	BA 40
	98	7.16	30	−64	53	Right Cerebrum	Parietal Lobe	Superior Parietal Lobule	BA 7
	19	5.82	48	−55	−16	Right Cerebrum	Temporal Lobe	Fusiform Gyrus	BA 37
	16	5.62	−42	5	29	Left Cerebrum	Frontal Lobe	Inferior Frontal Gyrus	
	13	5.14	−51	17	32	Left Cerebrum	Frontal Lobe	Middle Frontal Gyrus	

Table 2. Regions activated in post-stress recovery state in the high-stress group (control < stress, FWE 0.05).

Phase	Voxel	T-Value	x	y	z	Side	Lobe	Region	Brodmann Area
Stress	51	4.07	−60	−7	−16	Left Cerebrum	Temporal Lobe	Inferior Temporal Gyrus	BA 21
	31	2.48	−54	−19	−13	Left Cerebrum	Temporal Lobe	Middle Temporal Gyrus	
	171	3.60	0	−58	20	Left Cerebrum	Occipital Lobe	Precuneus	
	138	2.92	−6	−55	38	Left Cerebrum	Parietal Lobe	Precuneus	
	72	3.52	−18	−16	−19	Left Cerebrum	Limbic Lobe	Parahippocampal Gyrus	
	76	2.44	−9	41	2	Left Cerebrum	Limbic Lobe	Anterior Cingulate	BA 32
	77	2.41	18	44	−4	Right Cerebrum	Frontal Lobe	Medial Frontal Gyrus	BA 10
	68	3.15	−48	−70	26	Left Cerebrum	Temporal Lobe	Middle Temporal Gyrus	
	60	2.78	−51	−61	29	Left Cerebrum	Temporal Lobe	Middle Temporal Gyrus	
	58	2.71	−39	−67	35	Left Cerebrum	Parietal Lobe	Precuneus	BA 39
	33	2.98	48	−61	29	Right Cerebrum	Temporal Lobe	Middle Temporal Gyrus	
	30	2.64	51	−58	20	Right Cerebrum	Temporal Lobe	Superior Temporal Gyrus	
	15	2.84	60	2	−16	Right Cerebrum	Temporal Lobe	Middle Temporal Gyrus	BA 21
	12	2.54	−36	14	41	Left Cerebrum	Frontal Lobe	Middle Frontal Gyrus	
	9	1.74	−33	17	50	Left Cerebrum	Frontal Lobe	Superior Frontal Gyrus	
	14	2.51	−42	35	−7	Left Cerebrum	Frontal Lobe	Middle Frontal Gyrus	
	12	2.34	−48	32	2	Left Cerebrum	Frontal Lobe	Inferior Frontal Gyrus	
Recovery	1873	14.46	21	−97	−1	Right Cerebrum	Occipital Lobe	Cuneus	
	1807	13.96	−15	−97	−4	Left Cerebrum	Occipital Lobe	Lingual Gyrus	
	1803	13.93	30	−91	−10	Right Cerebrum	Occipital Lobe	Inferior Occipital Gyrus	
	203	8.11	−48	11	23	Left Cerebrum	Frontal Lobe	Inferior Frontal Gyrus	
	200	8.06	−45	35	23	Left Cerebrum	Frontal Lobe	Middle Frontal Gyrus	BA 9
	16	6.35	0	−55	−34	Left Cerebellum	Posterior Lobe	Cerebellar Tonsil	
	30	6.33	54	20	35	Right Cerebrum	Frontal Lobe	Middle Frontal Gyrus	BA 9
	29	6.20	42	8	29	Right Cerebrum	Frontal Lobe	Inferior Frontal Gyrus	
	25	5.85	−6	17	47	Left Cerebrum	Frontal Lobe	Medial Frontal Gyrus	BA 6
	22	5.42	3	14	50	Right Cerebrum	Frontal Lobe	Superior Frontal Gyrus	BA 6

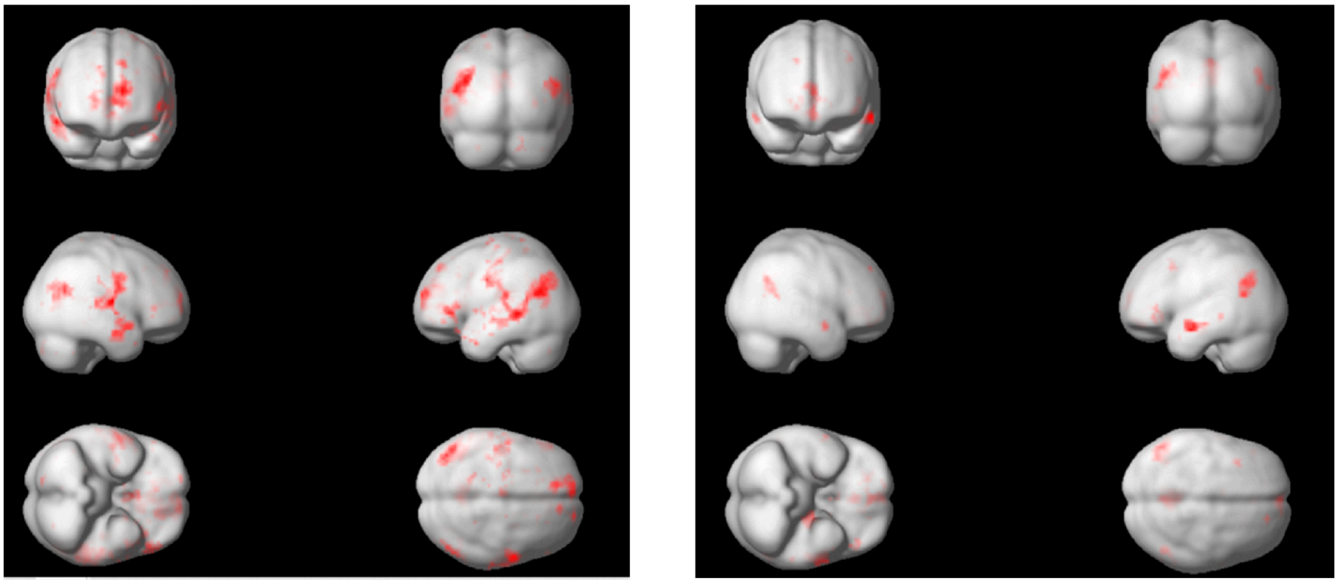


Figure 1. Brain mapping images of both the (**a**) moderate-stress and (**b**) high-stress groups during the stress-inducing condition.

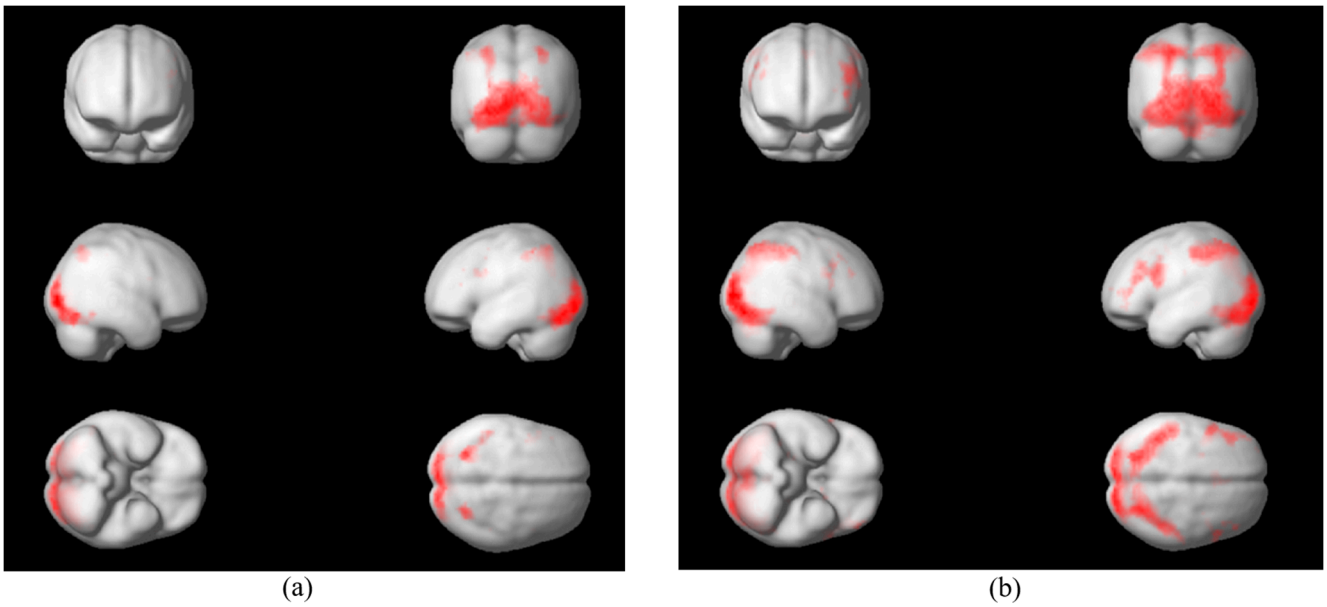


Figure 2. Brain mapping images of the two groups ((**a**) moderate-stress and (**b**) high-stress group) during the recovery state.

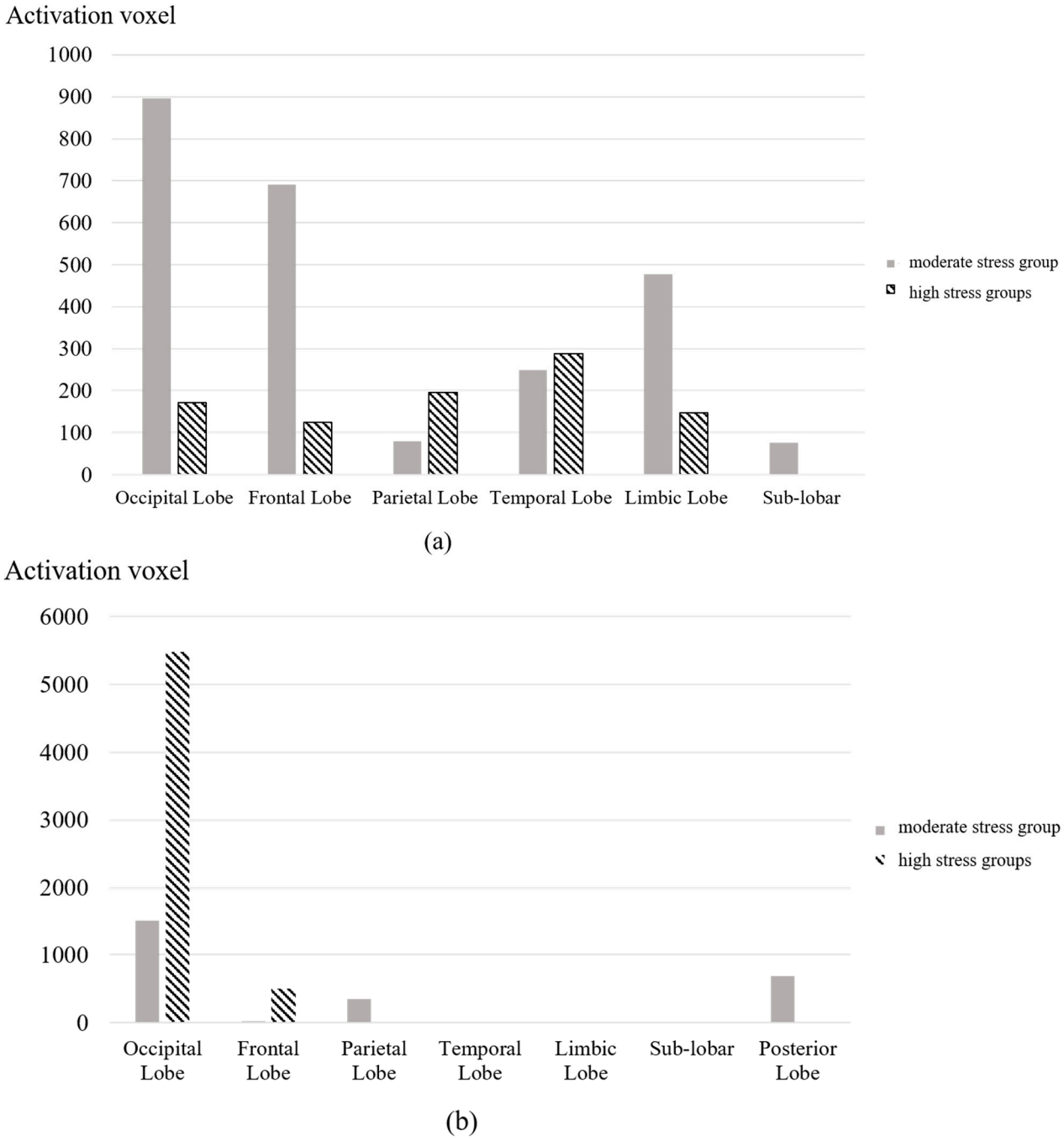


Figure 3. Comparison of activation voxels between the moderate- and high-stress groups during (**a**) stress-inducing conditions and (**b**) post-stress recovery state.

4. Discussion

This study identified and compared the activated brain areas and number of voxels in moderate- and high-stress groups using PSS scores under both stress-induced and recovery states.

Several brain regions are activated during stress-inducing tasks, including the amygdala, prefrontal cortex, anterior cingulate cortex, insula, and hypothalamus [11]. The amygdala, found to activate during acute and chronic stress, is involved in processing emotional information such as fear and anxiety [11]; the prefrontal cortex, activated during stress-related tasks, is associated with higher-order cognitive processes such as attention, decision making, and self-regulation [11]; the anterior cingulate cortex, involved in emotional

regulation, has been implicated in pain and anxiety perception [11]; the insula, activated during stress-related tasks, is linked to interoception and the processing of visceral sensations [11]; and finally, the hypothalamus region, also activated during stress-related tasks, is found to be involved in the regulation of the stress response [11].

Research on the relationship between stress and brain activation has found that, as stress levels increase, there is often increased activation in regions of the brain associated with the stress response, such as the amygdala, hypothalamus, and prefrontal cortex. Studies using various stressors, including the MIST, have consistently shown increased activation of these brain regions in response to stress [10,12]. Additionally, research has suggested that the strength of the relationship between stress and brain activation may depend on factors, such as sex, age, and history of stress exposure [13]. Overall, research suggests that changes in brain activation patterns are related to stress and may vary based on individual differences and the type of stressor experienced.

In this study, the areas of the brain activated in both the moderate- and high-stress groups when stress was induced were the middle temporal gyrus, superior frontal gyrus, anterior cingulate, inferior frontal gyrus, middle frontal gyrus, and parahippocampal gyrus (PHG) in the left hemisphere and the superior temporal gyrus and middle temporal gyrus in the right hemisphere. When comparing the number of activation voxels, the moderate-stress group showed a greater number of activation voxels than the high-stress group in most areas of the brain, while the reverse was true for the PHG in the limbic lobe, that is, the activation voxel number was greater in the high-stress group compared to the moderate-stress group. Functionally, the PHG engages in memory encoding, retrieval, and spatial navigation. While the PHG may not be the primary region involved in the stress response, one study found increased activation in the PHG during the MIST task compared with a control task, suggesting that it may play a role in the cognitive processing required for this task [10]. In individuals under high stress (compared with those under moderate stress), it is possible that increased activation of the PHG during stress-inducing tasks reflects a heightened emotional response to stress. This could be due to several factors, such as greater levels of stress-related hormones, heightened anxiety or fear, or greater attentional demands on emotional stimuli [4,14,15]. Additionally, the PHG is part of a large neural network involved in emotional processing, including the amygdala, prefrontal cortex, and hippocampus [14,16]. Increased activation of the PHG in individuals under high stress may reflect alterations in the connectivity or functional interactions between these regions, potentially resulting in a greater emotional response to stress.

Characteristic activation of the left precuneus was observed only in the high-stress group. The precuneus is a large region in the parietal lobe of the brain that is involved in a wide range of functions, including attention, memory, self-awareness, and the processing of emotional stimuli [17,18]. In the context of stress, the left precuneus has been implicated in the processing of emotional information and the regulation of negative affect [19,20]. Activation of the left precuneus during a stress-inducing task may reflect the participant's engagement in cognitive processing related to the task, such as attention, decision making, and emotional regulation. The left precuneus has been implicated in these types of processes, and its activation during stress-related tasks may reflect its role in the modulation of cognitive and emotional functions [19,20]. Moreover, the left precuneus is part of a network of brain regions that is activated during the processing of emotional information and the experience of negative affect [18]. This network includes regions such as the amygdala, insula, and prefrontal cortex, which, as previously described, are commonly activated during stress-related tasks. Overall, activation of the left precuneus during a stress-inducing task may reflect the participant's engagement in cognitive and emotional processing related to the task, as well as its role in a larger network of brain regions involved in the processing of emotional information and regulation of negative affect.

This study also compared the number of activation voxels in each lobe under stress-inducing conditions and showed that the moderate-stress group had a greater number of activation voxels in the occipital, frontal, and limbic lobes than the high-stress group, while

the reverse was true in the parietal lobe. It is difficult to determine the exact cause behind why the moderate-stress group showed greater activation in the occipital, frontal, and limbic lobes than the high-stress group based solely on the available information. There are several possible explanations for these findings. One possibility is that the moderate-stress group may have been more engaged with the task and, thus, showed greater activation in the brain regions associated with attention and cognitive control, such as the frontal lobe [1,15]. In contrast, the high-stress group may have been more preoccupied with their stressors and, thus, showed reduced engagement with the task, leading to lower levels of activation in these regions. Another possibility is that the moderate-stress group processed emotional stimuli differently from the high-stress group, leading to greater activation in the limbic lobe. This could be due to differences in the intensity or type of stress experienced by the two groups. This is supported by previous research showing that the amygdala and other limbic regions are involved in the processing of emotional stimuli and the regulation of affective responses [21,22]. Finally, the observed differences in brain activation between the moderate- and high- stress groups could be due to other factors, such as individual differences in stress reactivity, coping strategies, or the cognitive appraisal of stressors [23,24]. Overall, further research is needed to fully understand the factors that contribute to the differences in brain activation between individuals experiencing different levels of stress.

In the post-stress recovery phase, the left lingual, inferior frontal, and middle frontal gyri were activated in both groups, and the number of activated voxels in these common areas was greater in the high-stress group compared to the moderate-stress group. The finding that the left lingual, inferior frontal, and middle frontal gyri were commonly activated during the recovery phase after a stress task is consistent with previous research on brain activation patterns during post-stress recovery. These brain regions are involved in emotion regulation, attentional processes, and cognitive control, all of which are important for recovery from stress. However, the finding that the number of activation voxels in these areas was greater in the high-stress group compared to the moderate-stress group is a novel finding and may suggest that individuals with higher stress levels require more cognitive resources to regulate their emotions and recover from stress. One study consistent with our findings is that by Hermans et al. [25], who found that individuals with higher levels of stress showed increased activation in the inferior frontal gyrus during post-stress recovery, suggesting increased cognitive control during recovery.

In the moderate-stress group, the activation of the declive was characteristically observed in the left cerebellum. Some studies have investigated cerebellar activation during stress and recovery; however, these findings are inconsistent. Several studies have reported increased cerebellar activation during stress tasks, whereas others have reported decreases or no changes in activation [26]. Similarly, there are inconsistencies in the reported changes in cerebellar activation during recovery after stress. Some studies have reported increased cerebellar activation during recovery [26], whereas others have reported decreases or no changes in activation levels [27]. The declive is a cerebellum region reportedly involved in various cognitive and emotional processes, including working memory, attention, language processing, and emotional regulation. The activation of the declive during post-stress recovery may indicate that the brain is engaged in cognitive and emotional processing related to recovery from stress. Specifically, the greater activation of the declive in the moderate-stress group may suggest that this group engaged in more cognitive and emotional processing during recovery than the high-stress group, possibly reflecting more active efforts to cope with stressors and return to a baseline state. However, notably, the specific function (s) of the declive and its role in post-stress recovery are still not fully understood, and more research is needed to clarify these relationships.

In the high-stress group, the activation of the right cuneus was predominant. Previous studies have shown that the cuneus region is involved in visual processing, attention regulation, and the modulation of emotional responses. In particular, the cuneus plays a role in processing negative emotional stimuli, including those related to stress. Our

finding of greater activation of the right cuneus in the high-stress group during post-stress recovery is consistent with that of previous studies. For example, Veer et al. [28] found that individuals with high levels of perceived stress showed increased cuneus activation during an emotional processing task. Similarly, van Rooij et al. [5] found that stress exposure was associated with greater cuneus activation during an attentional control task. Taken together, our results suggest that the cuneus plays an important role in the recovery process after exposure to stress, particularly in individuals experiencing higher levels of stress.

While the activation of the inferior occipital gyrus was observed in both groups, this activation was predominantly observed in the left hemisphere in the moderate-stress group and in the right hemisphere in the high-stress group. The common activation of the inferior occipital gyrus in both the moderate- and high-stress groups during the recovery state after stress induction suggests that this area may play a role in regulating emotional and cognitive responses to stress. The lateralization of activation in the left and right inferior occipital gyri suggests differences in visual perception and attention processing between the two stress groups during stress recovery. Previous studies have reported the activation of the inferior occipital gyrus during stress and post-stress recovery. For example, Yang et al. [29] found that the inferior occipital gyrus is activated during stress induction and that the strength of this activation is positively correlated with subjective stress ratings. Another study by Kim et al. [11] reported increased activation of the inferior occipital gyrus during post-stress recovery in healthy adults.

5. Conclusions and Limitation

In this study, we examined the neural activation patterns during stress-inducing and post-stress recovery states in individuals classified into moderate- and high-stress groups using the Perceived Stress Scale (PSS). Our findings highlighted distinct differences in brain activation between the two groups. Under stress-inducing conditions, the moderate-stress group exhibited a greater number of activated voxels in the occipital, frontal, and limbic lobes, while the high-stress group showed predominant activation in the parietal lobe, particularly the left precuneus. During the recovery phase, these activation patterns shifted, with the high-stress group displaying more extensive voxel activation in the occipital and frontal lobes, suggesting heightened cognitive and emotional processing during recovery. Conversely, the moderate-stress group showed characteristic activation in the cerebellum, specifically the declive, implying distinct strategies for managing stress recovery.

The activation of specific brain regions, such as the anterior cingulate, prefrontal cortex, and parahippocampal gyrus, underscores their pivotal roles in emotional regulation, cognitive processing, and memory functions during stress and recovery. These findings suggest that the intensity and manner of stress responses can influence the brain's engagement in recovery processes. These data indicate that individuals with higher perceived stress may require more cognitive resources for emotional regulation, potentially impacting their recovery strategies.

The results of this study provide insight into the neural responses to stress and stress recovery, particularly in individuals with different levels of stress resilience. The activation of specific brain regions in response to stress and recovery highlights the importance of these regions in the regulation of stress responses and recovery. The findings of this study can be applied to the development of interventions or therapies aimed at improving stress resilience and promoting stress recovery. For example, targeting the specific activated brain regions identified in this study may be effective in reducing stress and promoting recovery in individuals with different levels of stress resilience.

Despite the insights gained from this study, several limitations should be acknowledged. First, the sample size, while sufficient for the current analysis, may limit the generalizability of the findings to the broader population. Future research with larger and more diverse participant groups is needed to strengthen the validity of these results. Second, the use of the Perceived Stress Scale (PSS) as the sole measure for stress classification, while widely accepted, is inherently subjective and may not capture the full complexity of

an individual's stress experience. Including objective stress biomarkers in future studies, such as cortisol levels, could provide a more comprehensive understanding of stress-related brain activation. Third, the cross-sectional design of this study restricts the ability to make causal inferences about the relationship between stress and brain activation. Longitudinal studies could offer more insights into how brain activation patterns change over time in response to varying stress levels. Additionally, the absence of significant activations in key subcortical regions like the amygdala and hippocampus may reflect limitations in spatial resolution, highlighting the need for targeted analyses in future research. Lastly, although efforts were made to control for potential confounding factors such as sleep patterns, diet, and physical activity, other unmeasured variables could have influenced the results. Addressing these limitations in future research will help to further clarify the neural mechanisms underlying stress and recovery.

Future research in this area could further investigate the neural mechanisms involved in stress and recovery among individuals with different levels of stress resilience. Furthermore, longitudinal studies might be conducted to examine how changes in brain activation patterns over time are related to changes in stress resilience and recovery. Further research could also examine the effectiveness of interventions that target the specific brain regions identified in this study to aid improvement in stress resilience and promote stress recovery.

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References

1. Arnsten, A.F. Stress signalling pathways that impair prefrontal cortex structure and function. *Nat. Rev. Neurosci.* **2009**, *10*, 410–422. [CrossRef] [PubMed]
2. McEwen, B.S. Brain on stress: How the social environment gets under the skin. *Proc. Natl. Acad. Sci. USA* **2012**, *109* (Suppl. S2), 17180–17185. [CrossRef] [PubMed]
3. Dedovic, K.; Duchesne, A.; Andrews, J.; Engert, V.; Pruessner, J.C. The brain and the stress axis: The neural correlates of cortisol regulation in response to stress. *NeuroImage* **2009**, *47*, 864–871. [CrossRef]
4. Liu, W.Z.; Zhang, W.H.; Zheng, Z.H.; Zou, J.X.; Liu, X.X.; Huang, S.H.; You, W.J.; He, Y.; Zhang, J.Y.; Wang, X.D.; et al. Identification of a prefrontal cortex-to-amygdala pathway for chronic stress-induced anxiety. *Nat. Commun.* **2020**, *11*, 2221. [CrossRef]
5. van Rooij, S.J.H.; Geuze, E.; Kennis, M.; Rademaker, A.R.; Vink, M. Neural activation during inhibition predicts improvement in PTSD symptoms: A randomized controlled trial of attention bias modification training. *NeuroImage Clin.* **2015**, *8*, 198–206. [CrossRef]
6. Elzinga, B.M.; Roelofs, K.; Tollenaar, M.S.; Bakvis, P.; van Pelt, J.; Spinhoven, P. Diminished cortisol responses to psychosocial stress associated with lifetime adverse events: A study among healthy young subjects. *Psychoneuroendocrinology* **2008**, *33*, 227–237. [CrossRef]
7. Gianaros, P.J.; Sheu, L.K.; Uyar, F.; Koushik, J.; Jennings, J.R.; Singh, T.D.W.T.D.A.; Verstynen, T.D. A brain phenotype for stressor-evoked blood pressure reactivity. *J. Am. Heart Assoc.* **2014**, *3*, e000524. [CrossRef]
8. Cohen, S.; Janicki-Deverts, D. Who's stressed? Distributions of psychological stress in the United States in probability samples from 1983, 2006, and 20091. *J. Appl. Soc. Psychol.* **2012**, *42*, 1320–1334. [CrossRef]
9. Lee, E.H. Review of the psychometric evidence of the perceived stress scale. *Asian Nurs. Res.* **2012**, *6*, 121–127. [CrossRef]

10. Dedovic, K.; Renwick, R.; Mahani, N.K.; Engert, V.; Lupien, S.P.; Pruessner, J.C. The Montreal Imaging Stress Task: Using functional imaging to investigate the effects of perceiving and processing psychosocial stress in the human brain. *J. Psychiatry Neurosci.* **2005**, *30*, 319–325.
11. Kim, M.J.; Loucks, R.A.; Palmer, A.L.; Brown, A.C.; Solomon, K.M.; Marchante, A.N.; Whalen, P.J. The structural and functional connectivity of the amygdala: From normal emotion to pathological anxiety. *Behav. Brain Res.* **2011**, *223*, 403–410. [CrossRef] [PubMed]
12. Gianaros, P.J.; Jennings, J.R.; Sheu, L.K.; Greer, P.J.; Kuller, L.H.; Matthews, K.A. Prospective reports of chronic life stress predict decreased gray matter volume in the hippocampus. *NeuroImage* **2008**, *35*, 795–803. [CrossRef] [PubMed]
13. Gianaros, P.J.; Wager, T.D. Wager Brain-body pathways linking psychological stress and physical health. *Curr. Dir. Psychol. Sci.* **2015**, *24*, 313–321. [CrossRef] [PubMed]
14. Salzman, C.D.; Fusi, S. Emotion; cognition, and mental state representation in amygdala and prefrontal cortex. *Annu. Rev. Neurosci.* **2010**, *33*, 173–202. [CrossRef] [PubMed]
15. Eysenck, M.W.; Derakshan, N.; Santos, R.; Calvo, M.G. Anxiety and cognitive performance: Attentional control theory. *Emotion* **2007**, *7*, 336–353. [CrossRef]
16. Šimić, G.; Tkalčić, M.; Vukić, V.; Mulc, D.; Španić, E.; Šagud, M.; Olucha-Bordonau, F.E.; Vukšić, M.; Hof, P.R. Understanding Emotions: Origins and Roles of the Amygdala. *Biomolecules* **2021**, *11*, 823. [CrossRef]
17. Cavanna, A.E.; Trimble, M.R. The precuneus: A review of its functional anatomy and behavioural correlates. *Brain* **2006**, *129*, 564–583. [CrossRef]
18. Margulies, D.S.; Vincent, J.L.; Kelly, C.; Lohmann, G.; Uddin, L.Q.; Biswal, B.B.; Villringer, A.; Castellanos, F.X.; Milham, M.P.; Petrides, M. Precuneus shares intrinsic functional architecture in humans and monkeys. *Proc. Natl. Acad. Sci. USA* **2009**, *106*, 20069–20074. [CrossRef]
19. Dedovic, K.; Rexroth, M.; Wolff, E.; Duchesne, A.; Scherling, C.; Beaudry, T.; Lue, S.D.; Engert, C.L.V.; Pruessner, J.C. Neural correlates of processing stressful information: An event-related fMRI study. *Brain Res.* **2009**, *1293*, 49–60. [CrossRef]
20. Wang, Y.; Qin, W.; Liu, B.; Wang, D.; Zhang, Y.; Jiang, T.; Lin, C.P. Anatomical insights into disrupted small-world networks in schizophrenia. *NeuroImage Clin.* **2012**, *16*, 1085–1093. [CrossRef]
21. Etkin, A.; Wage, T.D. Functional neuroimaging of anxiety: A meta-analysis of emotional processing in PTSD, social anxiety disorder, and specific phobia. *Am. J. Psychiatry.* **2007**, *164*, 1476–1488. [CrossRef] [PubMed]
22. Phelp, E.A.; LeDoux, J.E. Contributions of the amygdala to emotion processing: From animal models to human behavior. *Neuron* **2005**, *48*, 175–187. [CrossRef] [PubMed]
23. Lazarus, R.S.; Folkman, S. *Stress, Appraisal, and Coping*; Springer Publishing Company: Berlin/Heidelberg, Germany, 1984.
24. Mak, A.K.; Hu, Z.G.; Zhang, J.X.; Xiao, Z.W.; Lee, T.M. Neural correlates of regulation of positive and negative emotions: An fmri study. *Neurosci. Lett.* **2009**, *457*, 101–106. [CrossRef] [PubMed]
25. Hermans, E.J.; van Marle, H.J.; Ossewaarde, L.; Henckens, M.J.; Qin, S.; van Kesteren, M.T.; Schoots, V.C.; Cousijn, H.; Rijpkema, M.; Oostenveld, R.; et al. Stress-related noradrenergic activity prompts large-scale neural network reconfiguration. *Science* **2011**, *334*, 1151–1153. [CrossRef]
26. van Oort, J.; Tendolkar, I.; Hermans, E.J.; Mulders, P.C.; Beckmann, C.F.; Schene, A.H.; Fernández, G.; van Eijndhoven, P.F. How the brain connects in response to acute stress: A review at the human brain systems level. *Neurosci. Biobehav. Rev.* **2017**, *83*, 281–297. [CrossRef]
27. Sambataro, F.; Wolf, N.D.; Pennuto, M.; Vasic, N.; Wolf, R.C. Revisiting default mode network function in major depression: Evidence for disrupted subsystem connectivity. *Psychol. Med.* **2014**, *44*, 2041–2051. [CrossRef]
28. Veer, I.M.; Oei, N.Y.; Spinhoven, P.; van Buchem, M.A.; Elzinga, B.M.; Rombouts, S.A. Beyond acute social stress: Increased functional connectivity between amygdala and cortical midline structures. *NeuroImage* **2011**, *57*, 1534–1541. [CrossRef]
29. Fan, S.; Yu, Y.; Wu, Y.; Kai, Y.; Wang, H.; Chen, Y.; Zu, M.; Pang, X.; Tian, Y. Altered brain entropy and functional connectivity patterns in generalized anxiety disorder patients. *J. Affect. Disord.* **2023**, *332*, 168–175. [CrossRef]

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